

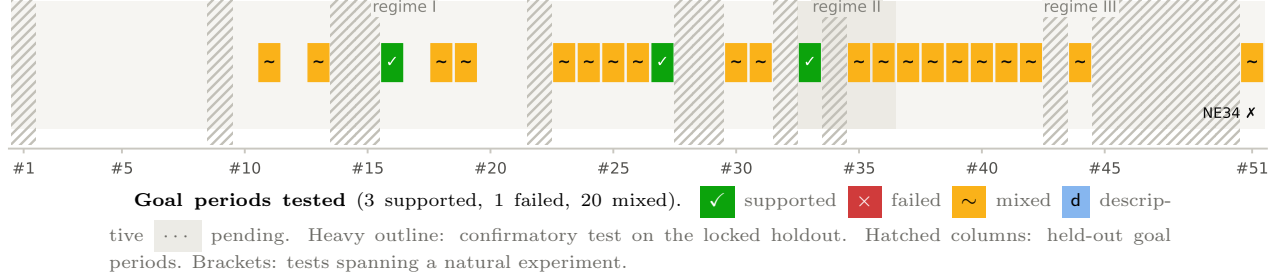
H12: Groupthink is dimensional collapse: few collective modes, shrinking participation ratio at consensus

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. The swarm has one collective mode and does not collapse at kickoffs; its low-dimensional episodes are agents looping on themselves, which a cheap self-repetition rate detects.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

24 non-holdout units (goal periods split at step changes), two arms. **Spectrum:** the agent×agent correlation matrices of activity, talk and chat content each have exactly one eigenvalue above the cross-day surrogate edge (22/24, 20/24, 24/24 units), never more than two. It is uniform across agents (VR/ λ_1 median 0.97): the Curie–Weiss “market mode”, ranking like H02’s βJ_0 ($\rho = 0.73$). But it is partly synchronized lulls: it survives dropping minutes with ≤ 1 active agent in only 14/24 units. **Dimensionality failed, in direction:** the participation ratio of chat did not collapse at 20 kickoffs (median +7%, $p = 0.86$ vs placebos; +44% in regime III); day 1 is more diverse than later days (13/16, $p = 0.02$); free weeks are not more diverse than shared ones ($p = 0.58$). The lowest-PR weeks (#38–#40) are agents repeating their own text (16–60% self near-copies), not copying each other. Hold-out script frozen, not run.

PHYSICAL MODEL

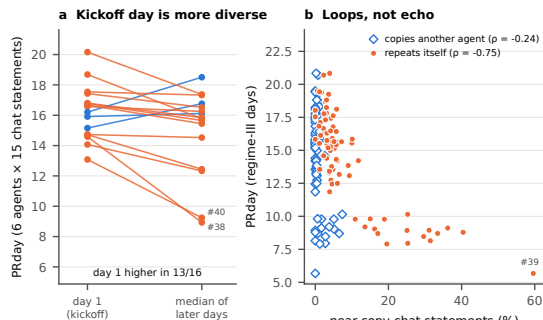
Models 01 (inverse Ising), 11 (vector spins), random-matrix theory. $s_i(t)$: agent i ’s ± 1 activity or talk spin per minute, or whitened content vector per 30-min window (N agents, T samples). C : equal-time correlation matrix, eigenvalues λ_k ; λ_+ : noise edge (used: cross-day surrogates); $u = \mathbf{1}/\sqrt{N}$; $\bar{\rho}$: mean pair correlation. PR: participation ratio of the content covariance Σ ($d = 32$), bias-corrected via $W = \nu S$, $\nu = n - 1$.

$$C_{ij} = \langle s_i s_j \rangle_t, \quad \lambda_+ = (1 + \sqrt{N/T})^2, \quad k = \#\{\lambda_k > \lambda_+\}$$

$$\text{VR} = u^\top C u \leq \lambda_1 \approx 1 + (N - 1)\bar{\rho} \quad (\text{iff uniform mode})$$

$$\text{PR} = (\text{tr } \Sigma)^2 / \text{tr } \Sigma^2 = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$$

$$\mathbb{E}(\text{tr } W)^2 = \nu^2 a + 2\nu b, \quad \mathbb{E} \text{tr } W^2 = \nu a + \nu(\nu+1)b \Rightarrow \hat{a}, \hat{b}$$



Observables. (a) Agent-balanced PR (6 agents × 15 chat statements) on the kickoff day vs the median of later days, for the 16 periods with $N \geq 10$: day 1 is higher in 13 (orange), the opposite of a kickoff collapse. (b) Regime-III days: PR falls as agents near-copy (cosine > 0.95) their own earlier messages ($\rho = -0.75$), not other agents’ ($\rho = -0.24$; under 8% on every day). The low cluster is #38–#40. Post hoc; decides no verdict.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The most portable result refutes its own premise. Dimensional collapse is a bad groupthink alarm: kickoffs make agents *more* diverse, and the truly low-dimensional states were single agents stuck in loops, re-posting near-copies of their own messages. Anyone with an (agent, time, text) log can compute self-repetition and cross-agent echo rates in minutes, with nothing fitted; self-repetition flagged this village’s worst weeks. That is a cheap stuck-agent monitor. The “market mode” is less useful: agents are busy or idle together, partly platform-wide lulls. The groupthink alarm stays unvalidated: echo never exceeded 8% here.

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